

14.3 The Four Types of SVD



The core skills you'll master in this section:

- ▶ Full SVD: U and V are both orthogonal matrices, and S has the same shape as the original matrix.
- ▶ Economy SVD: makes S a diagonal square matrix, and U a semi-orthogonal matrix.
- ▶ Compact SVD: only when a tall-and-thin matrix is not full column rank can the economy SVD be compressed into the compact SVD, removing the parts with singular value 0.
- ▶ Truncated SVD: takes the first p principal components to approximately reconstruct the original matrix – a common technique for data compression.

The first section of this chapter mentioned that there are four types of singular value decomposition:

- ▶ Full;
- ▶ Economy-size (thin);
- ▶ Compact;
- ▶ Truncated.

This section will introduce these four types of singular value decomposition in depth.



Before reading this section, we recommend reviewing block matrix multiplication in Chapter 3, Section 5.

Full: S has the same shape as X

Given an arbitrary real matrix $X_{n \times D}$, performing full singular value decomposition (SVD) yields

$$X_{n \times D} = U_{n \times n} S_{n \times D} V_{D \times D}^T$$

As shown in Figure 1, in the full SVD, U and V are both orthogonal matrices, and S 's shape is the same as X 's.

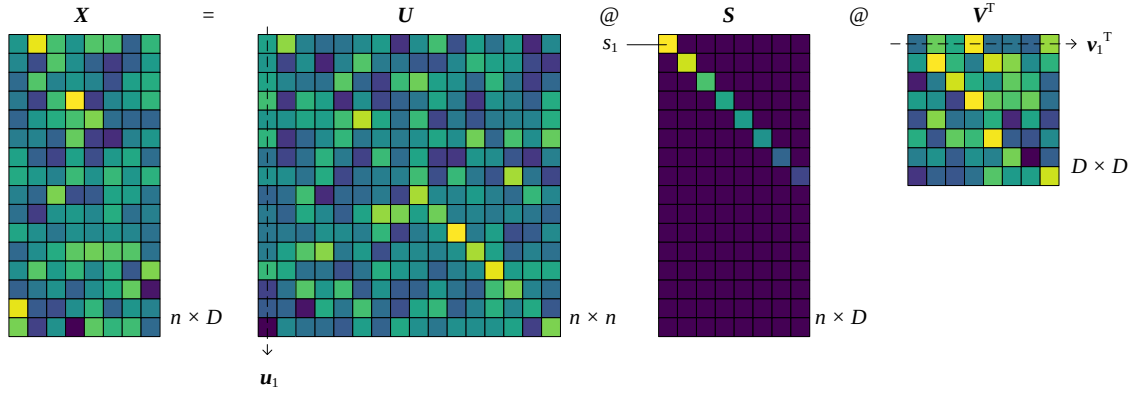


Figure 1. Full SVD

Economy: S removes the zero matrix and becomes a square matrix

Building on the full SVD, the rectangular diagonal matrix $\mathbf{S}_{n \times D}$ is partitioned vertically into a diagonal square matrix and a zero matrix, $\mathbf{O}$:

$$\mathbf{S}_{n \times D} = \begin{bmatrix} s_1 & & & \\ & s_2 & & \\ & & \ddots & \\ & & & s_D \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} = \begin{bmatrix} \mathbf{S}_{D \times D} \\ \mathbf{O}_{(n-D) \times D} \end{bmatrix}$$

Write $\mathbf{U}_{n \times n}$ as the left-right block matrix $[\mathbf{U}_{n \times D}, \mathbf{U}_{n \times (n-D)}]$, where $\mathbf{U}_{n \times D}$ and $\mathbf{X}$ have the same shape.

As shown in Figure 2, using block matrix multiplication, the full SVD can be simplified into the economy SVD:

$$\begin{aligned} \mathbf{X}_{n \times D} &= [\mathbf{U}_{n \times D} \quad \mathbf{U}_{n \times (n-D)}] \begin{bmatrix} \mathbf{S}_{D \times D} \\ \mathbf{O}_{(n-D) \times D} \end{bmatrix} \mathbf{V}^T \\ &= (\mathbf{U}_{n \times D} \mathbf{S}_{D \times D} + \mathbf{U}_{n \times (n-D)} \mathbf{O}_{(n-D) \times D}) \mathbf{V}^T \\ &= \mathbf{U}_{n \times D} \mathbf{S}_{D \times D} \mathbf{V}^T \end{aligned}$$

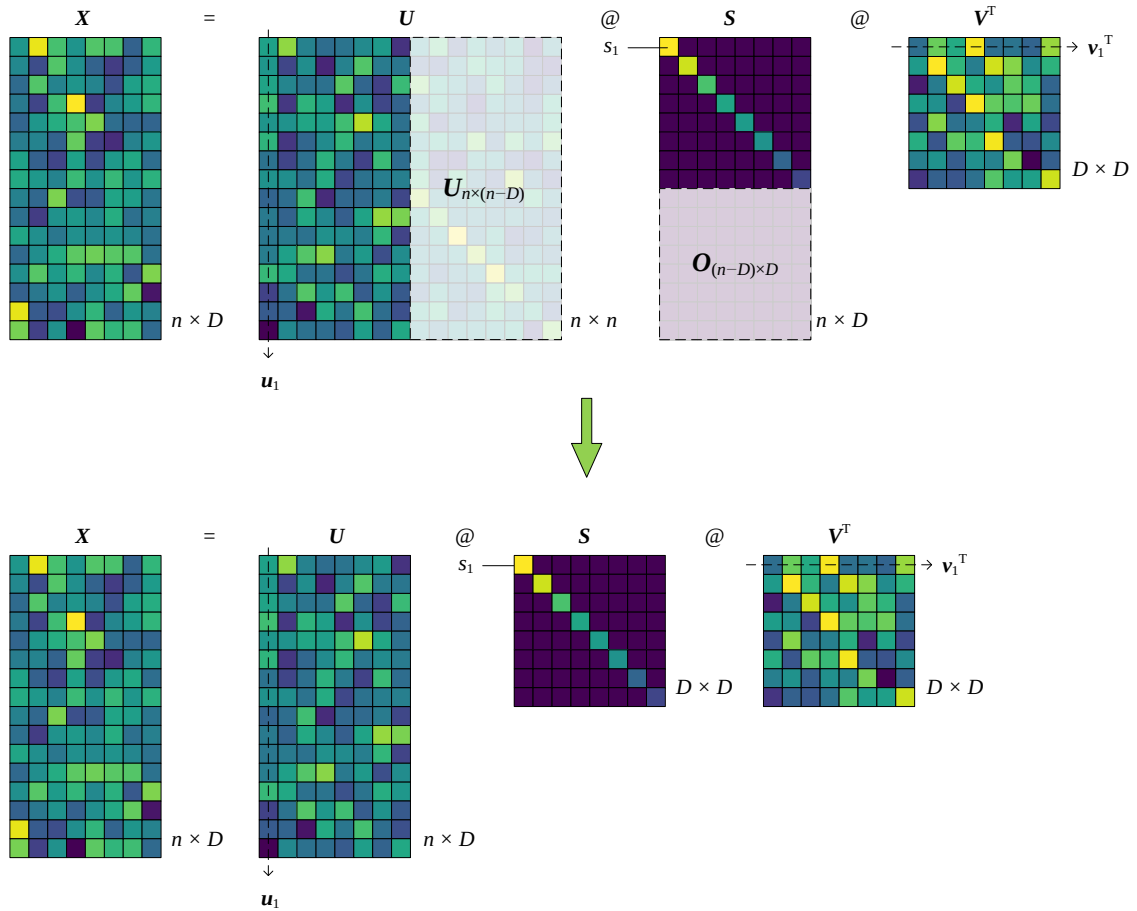
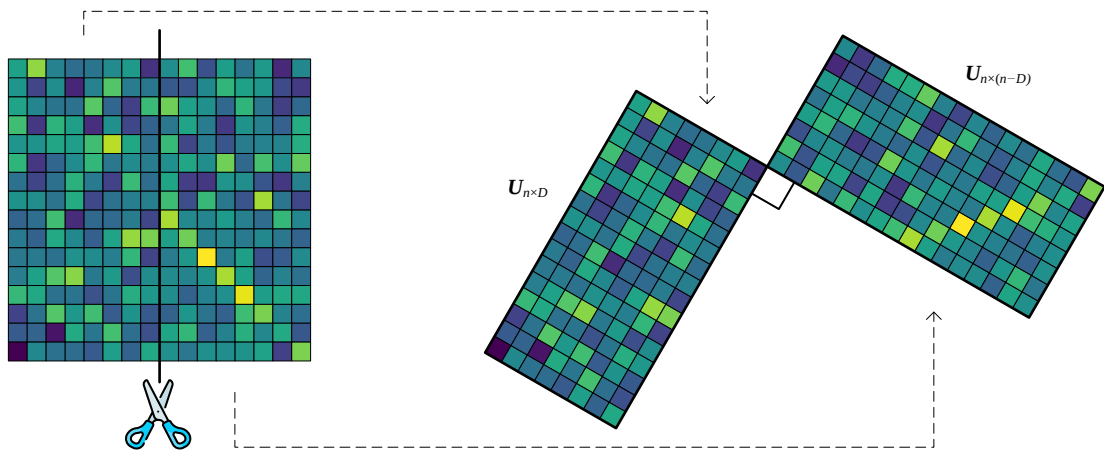
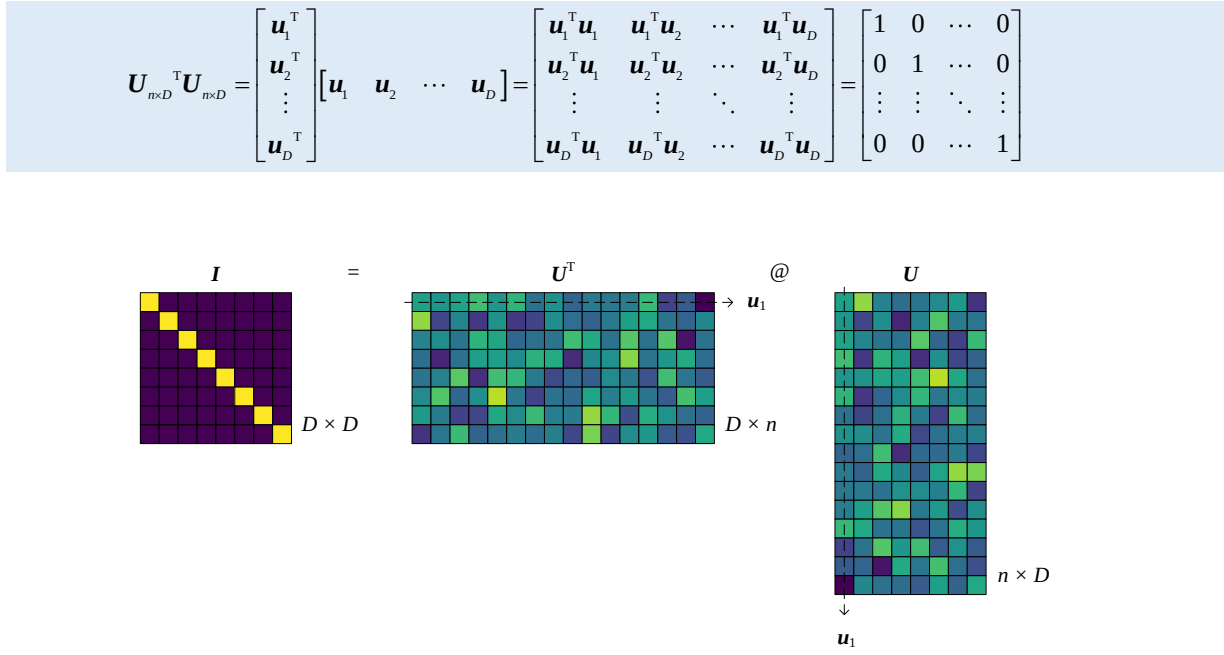


Figure 2. SVD, from Full to Economy, X is Full Column Rank

As shown in Figure 3, $U_{n \times D}$ and $U_{n \times (n-D)}$'s column vectors span spaces that are orthogonal complements of each other.

Figure 3. $U_{n \times D}$ and $U_{n \times (n-D)}$'s Column Vectors Span Orthogonal Complement Spaces

$U_{n \times D}$ is no longer an orthogonal matrix, but rather a semi-orthogonal matrix. As shown in Figure 4, $U_{n \times D}^T$ and $U_{n \times D}$'s matrix product is still the identity matrix, that is,

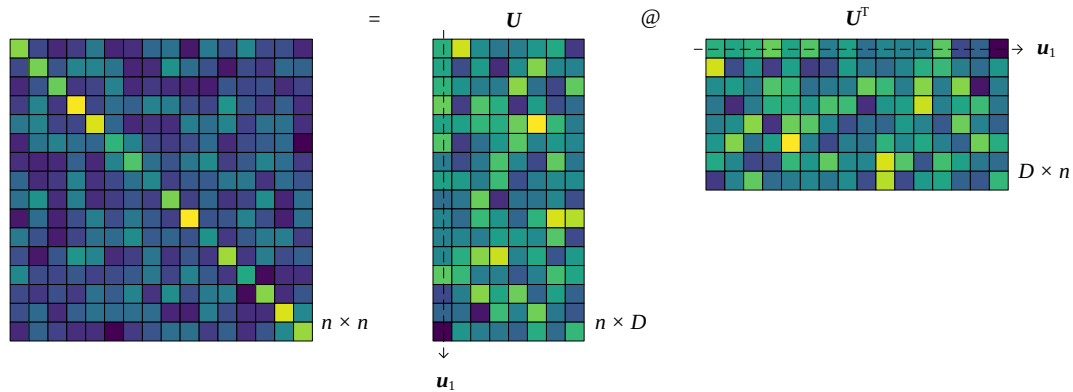
Figure 4. $U_{n \times D}^T$ and $U_{n \times D}$'s Matrix Product is Still the Identity Matrix

But, as shown in Figure 5, $U_{n \times D}$ and $U_{n \times D}^T$'s product is no longer the identity matrix. Expanding $U_{n \times D} @ U_{n \times D}^T$ gives

$$U_{n \times D} U_{n \times D}^T = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_D \end{bmatrix} \begin{bmatrix} \mathbf{u}_1^T \\ \mathbf{u}_2^T \\ \vdots \\ \mathbf{u}_D^T \end{bmatrix} = \mathbf{u}_1 \mathbf{u}_1^T + \mathbf{u}_2 \mathbf{u}_2^T + \cdots + \mathbf{u}_D \mathbf{u}_D^T = I - (\mathbf{u}_{D+1} \mathbf{u}_{D+1}^T + \mathbf{u}_{D+2} \mathbf{u}_{D+2}^T + \cdots + \mathbf{u}_n \mathbf{u}_n^T)$$

? Please analyze the equation above from the perspective of an orthogonal projection matrix.

➔ Please review the content on orthogonal matrices in the first three sections of Chapter 9 of this book.

Figure 5. $U_{n \times D}$ and $U_{n \times D}^T$'s Product is No Longer the Identity Matrix

Comparing Figure 2, full SVD and economy SVD — the only thing that stays unchanged in the decomposition result is matrix V , which always remains a square matrix.

Flat Matrix

Please note that a flat matrix's full SVD (Figure 6) and economy SVD (Figure 7) differ from those of a tall-and-thin matrix.

? Please analyze Figure 6 and Figure 7 on your own.

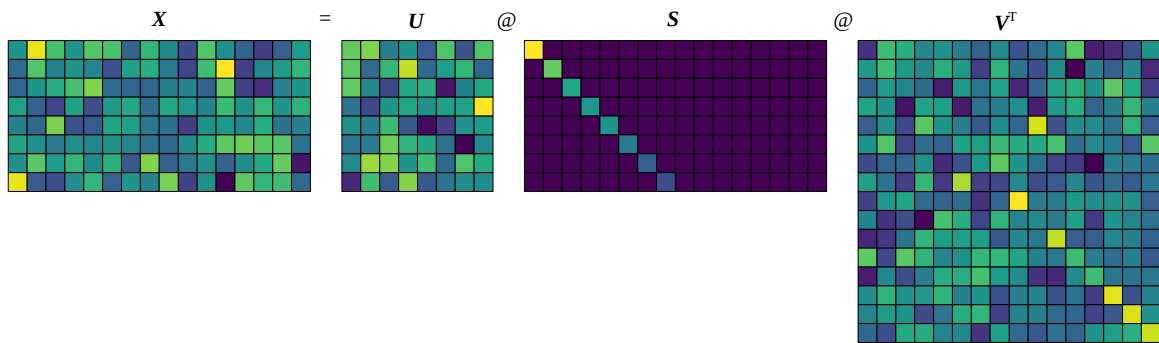


Figure 6. Full SVD, Flat Matrix

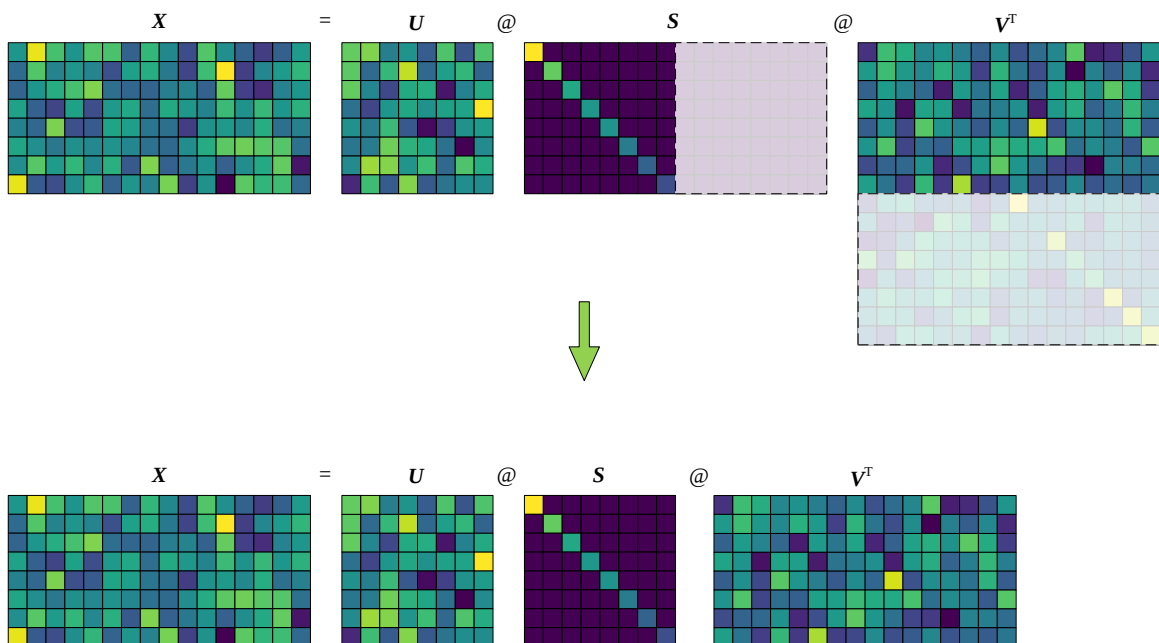


Figure 7. SVD, from Full to Economy, Flat Matrix

Compact: Tall-and-Thin Matrix Not Full Column Rank

Next, let's see how to obtain the compact SVD building on the economy SVD.

! Note that the compact SVD exists only when a tall-and-thin matrix is not full column rank.

In particular, if $\text{rank}(\mathbf{X}) = r < D$, the singular values s_j satisfy:

$$s_1 \geq s_2 \geq \dots s_r > 0, \quad s_{r+1} = s_{r+2} = \dots s_D = 0$$

Under this condition, the singular value square matrix $\mathbf{S}$ obtained from the economy SVD can be split into four sub-blocks:

$$\mathbf{S} = \begin{bmatrix} \mathbf{S}_{r \times r} & \mathbf{O}_{r \times (D-r)} \\ \mathbf{O}_{(D-r) \times r} & \mathbf{O}_{(D-r) \times (D-r)} \end{bmatrix}$$

In the equation above, matrix $\mathbf{S}_{r \times r}$'s diagonal elements – the singular values – are all greater than 0.

Substituting the equation above into the economy SVD, and simplifying, we get:

$$\begin{aligned} \mathbf{X}_{n \times D} &= \begin{bmatrix} \mathbf{U}_{n \times r} & \mathbf{U}_{n \times (D-r)} \end{bmatrix} \begin{bmatrix} \mathbf{S}_{r \times r} & \mathbf{O}_{r \times (D-r)} \\ \mathbf{O}_{(D-r) \times r} & \mathbf{O}_{(D-r) \times (D-r)} \end{bmatrix} \begin{bmatrix} \mathbf{V}_{D \times r} & \mathbf{V}_{D \times (D-r)} \end{bmatrix}^T \\ &= \begin{bmatrix} \mathbf{U}_{n \times r} \mathbf{S}_{r \times r} & \mathbf{O}_{n \times (D-r)} \end{bmatrix} \begin{bmatrix} (\mathbf{V}_{D \times r})^T \\ (\mathbf{V}_{D \times (D-r)})^T \end{bmatrix} \\ &= \mathbf{U}_{n \times r} \mathbf{S}_{r \times r} (\mathbf{V}_{D \times r})^T \end{aligned}$$

Please pay special attention that, in the equation above, matrix $\mathbf{V}$ is first partitioned and then transposed.

⚠ Note that in the compact SVD, $\mathbf{U}$ and $\mathbf{V}$ are neither of them square matrices.

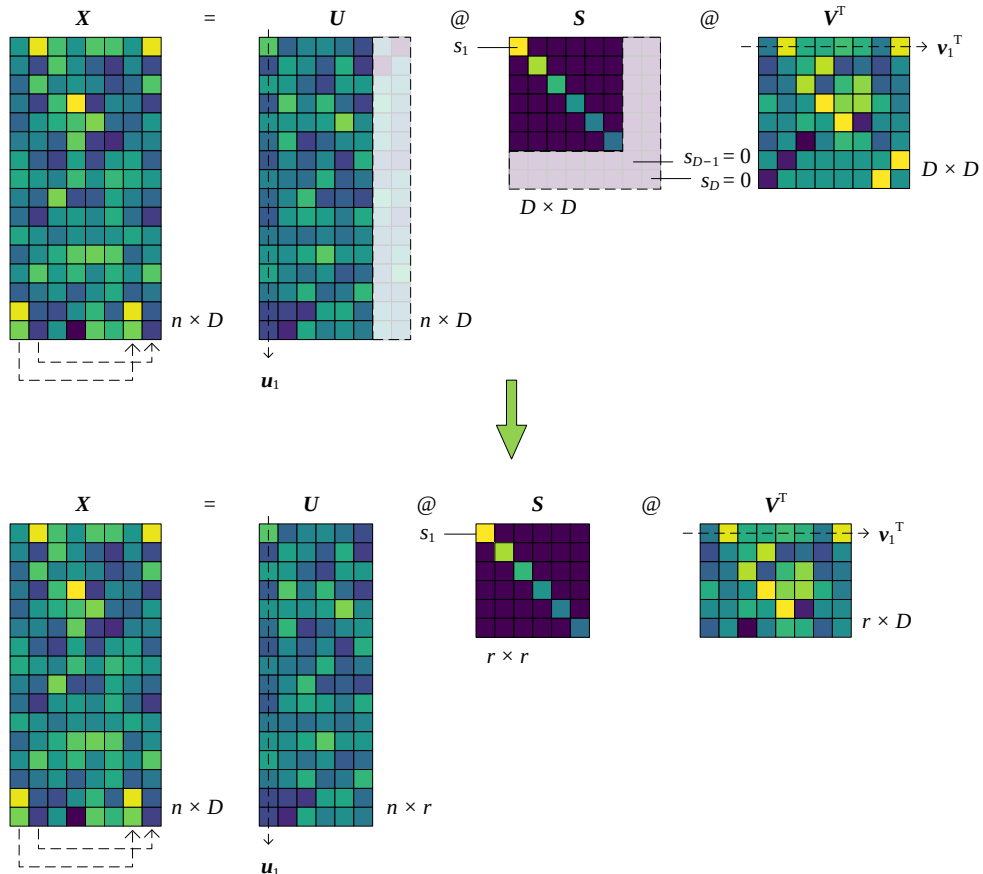


Figure 8. SVD, from Economy to Compact, X is Not Full Column Rank

It's especially worth noting that, after a matrix undergoes singular value decomposition, the number of nonzero singular values corresponds to the matrix's column rank and row rank.

A matrix's column rank is the maximum number of linearly independent vectors among all its column vectors; its row rank is the maximum number of linearly independent vectors among all its row vectors.

Figure 8 tells us that U 's column vectors span the same space as X 's column vectors; V^T 's row vectors span the same space as X 's row vectors. The number of nonzero singular values determines the dimension of that space – that is, the matrix's rank, column rank, and row rank.

Truncated: Approximation

If $\text{rank}(X) = r \leq D$, take the matrix components corresponding to the first p singular values ($p < r$) in the economy SVD – U , S , and V – and use them to reconstruct the original data; this is the truncated SVD:

$$X_{n \times D} \approx \hat{X}_{n \times D} = U_{n \times p} S_{p \times p} (V_{D \times p})^T$$

Please fill in the matrix partitioning and corresponding multiplication in the equation above on your own.

The symbol in the equation above is not an equals sign – meaning the truncated SVD cannot fully recover the original data. In other words, the truncated SVD is an approximation of the original matrix X .

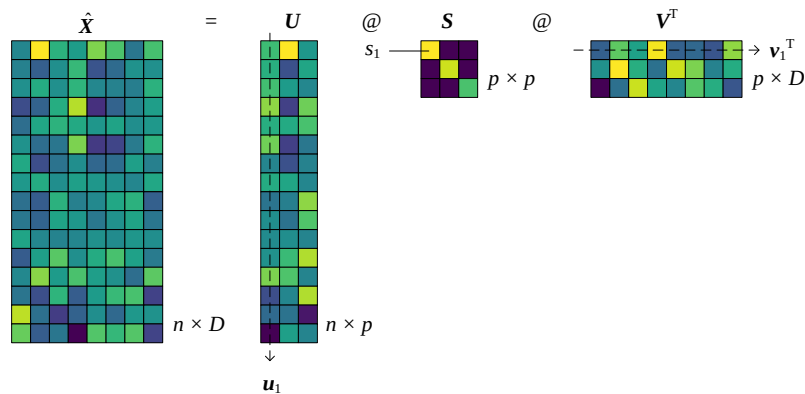


Figure 9. Approximating the Original Data Using Truncated SVD

Figure 9 shows the heatmaps of the truncated SVD; we can see that $X_{n \times D}$ and $\hat{X}_{n \times D}$ – the two heatmaps show a certain "color difference", that is, the difference between the following two (identically shaped) matrices

$$X_{\text{error}} = X_{n \times D} - \hat{X}_{n \times D}$$

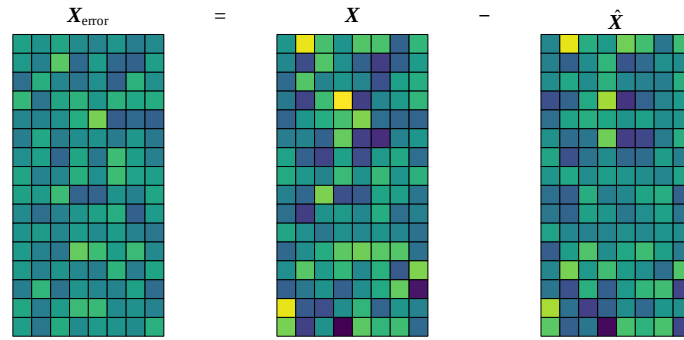


Figure 10. Error

Where does this error come from? That's the topic of the next section!



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please perform the full singular value decomposition of the following matrix, and then write the result as an economy SVD.

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$$

Q2. Please write the full SVD, economy SVD, and compact SVD of the following matrix.

$$A = \begin{bmatrix} 0 & 0 \\ 1 & -1 \\ 1 & -1 \end{bmatrix}$$

Q3. Compute the row rank and column rank of the matrix A in Q1 and Q2, respectively.

Q4. For the matrices in Q1 and Q2, respectively compute matrix A 's Gram matrix $A^T A$, as well as $A^T A$'s row rank and column rank.

Q5. For the matrices in Q1 and Q2, respectively compute matrix A^T 's Gram matrix AA^T , as well as AA^T 's row rank and column rank.